Operating Systems – 3 easy pieces

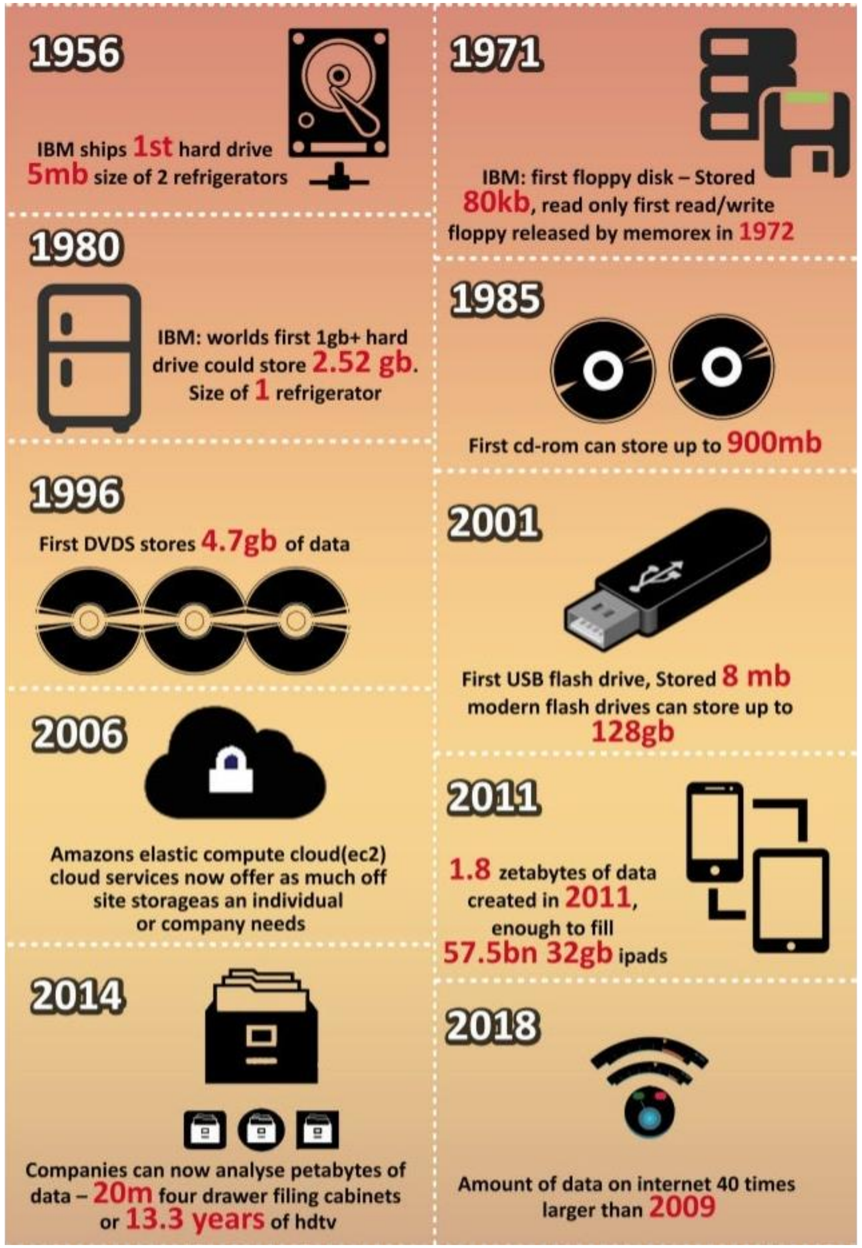
Chap. 37: Hard disk drives

Dr. B. Wajid

Monday: Sep. 2, 2024


I don't think you even know what a hard drive is.
I've driven cross country with my wife, three kids and a dog. I know what a hard drive is.
www.pak101.com

- Inspiration: It is worth *understanding the details of a disk's operation* before *building the file system software that manages it*. Therefore, these next two chapters, 37 and 38, talk about disk drives.


1956
IBM ships 1st hard drive
5mb size of 2 refrigerators
1971
IBM: first floppy disk – Stored
80kb, read only first read/write
floppy released by memorex in 1972
1980
IBM: worlds first 1gb+ hard
drive could store 2.52 gb.
Size of 1 refrigerator
1985
First cd-rom can store up to 900mb
1996
First DVDS stores 4.7gb of data
2001
First USB flash drive, Stored 8 mb
modern flash drives can store up to
128gb
2006
Amazons elastic compute cloud(ec2)
cloud services now offer as much off
site storageas an individual
or company needs
2011
1.8 zetabytes of data
created in 2011,
enough to fill
57.5bn 32gb ipads
2014
Companies can now analyse petabytes of
data – 20m four drawer filing cabinets
or 13.3 years of hdtv
2018
Amount of data on internet 40 times
larger than 2009

Hard disk Driver

Hard disk driver are the **main form of persistent data storage** in computers.

- The drive consists of **sectors** (512-byte blocks).
- **Atomic**: When updating the disk, drive manufacturers can only guarantee making a single 512-byte write.
- It will either write or won't write at all (called a **torn write**).
- Accessing blocks near to each other is faster (**sequential read/write**) than accessing blocks far away from each other (**called random access**).
- **Address Space:** We can view the disk with n sectors as an array of sectors; $\mathbf{0} \rightarrow (\boldsymbol{N}-\mathbf{1})$.

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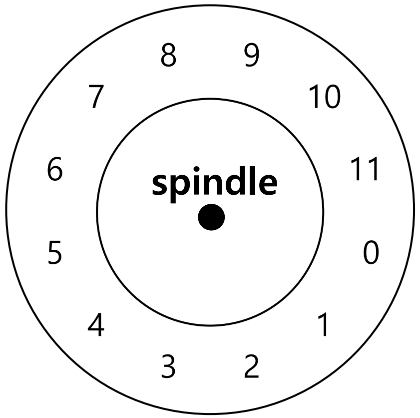

A Disk with Just A Single Track (12 sectors)

- **Platter** (Aluminum coated with a thin magnetic layer)
- A circular hard surface. Data is stored by inducing magnetic changes to it.
- Each platter has 2 sides, each of which is called a **surface**.
- The platters are all bound together around the **spindle**.

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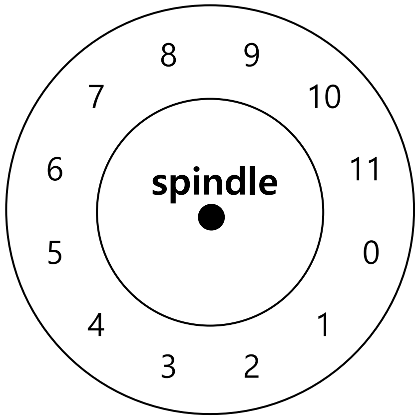

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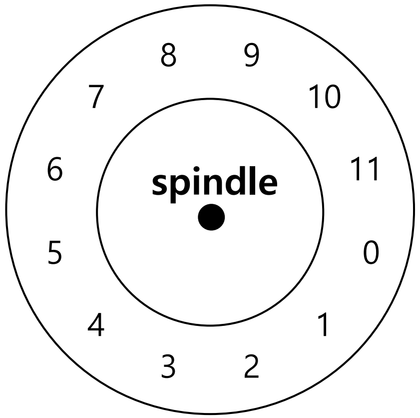

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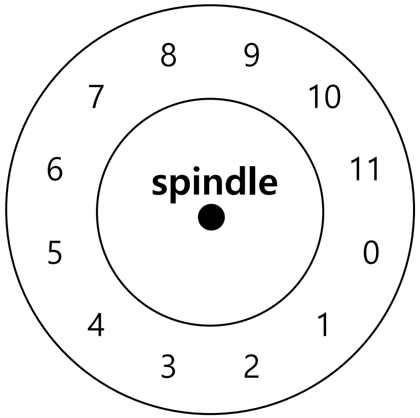

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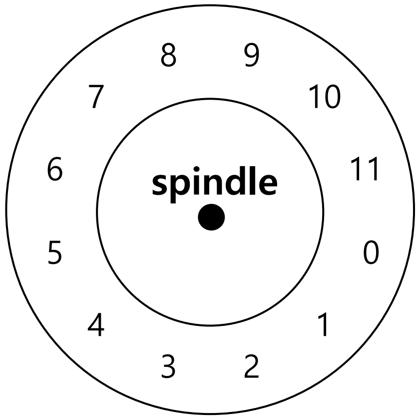

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- The platters are all bound together around the **spindle**.
- Spindle is **connected to a motor** that **spins** the platters around.
- The rate of rotations is measured in **RPM** (Rotations Per Minute). Typical modern values : 7,200 RPM to 15,000 RPM.
- E.g., 10000 RPM : A single rotation takes about 6 ms.

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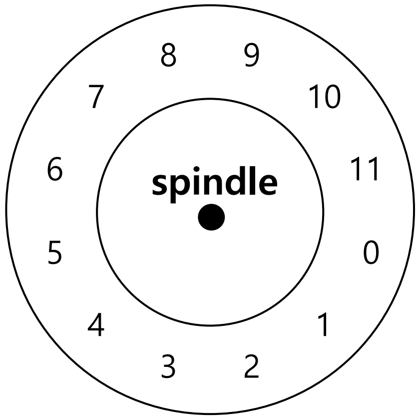

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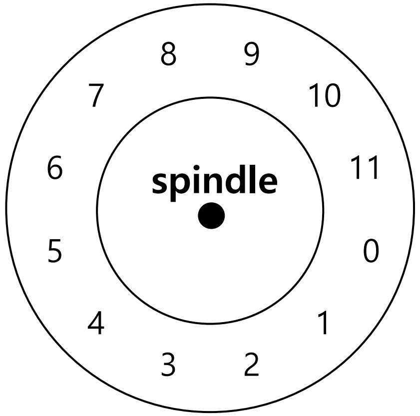

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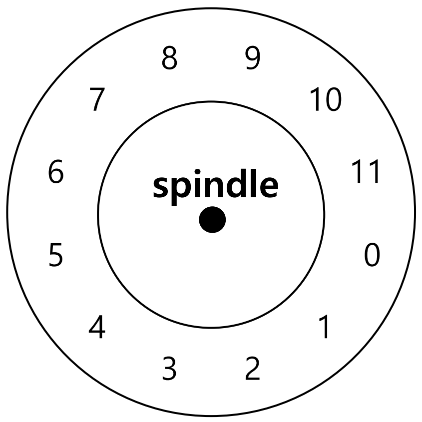

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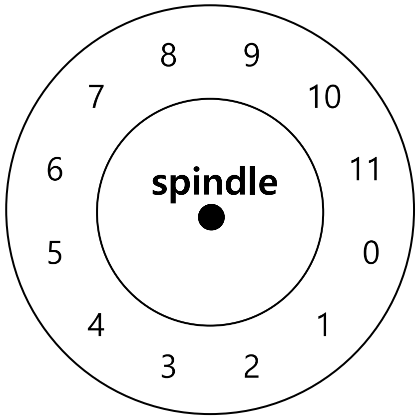

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- A single surface contains many thousands and thousands of tracks.

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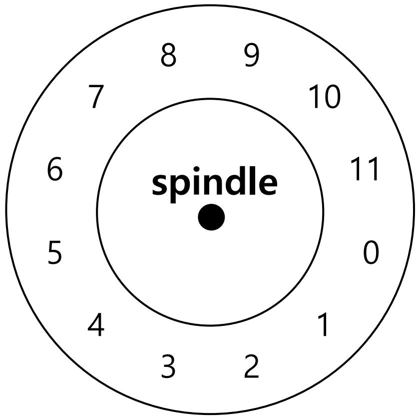

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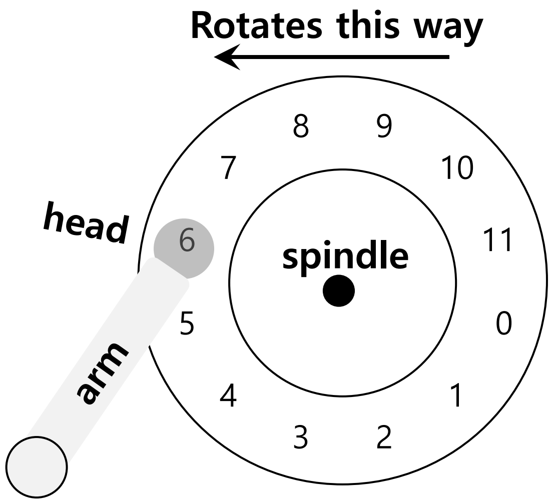

A Single Track Plus A Head

- **Disk head** (One head per surface of the drive)
- The process of ***reading*** and ***writing*** is accomplished by the **disk head**.
- Attached to a single **disk arm**, which moves across the surface

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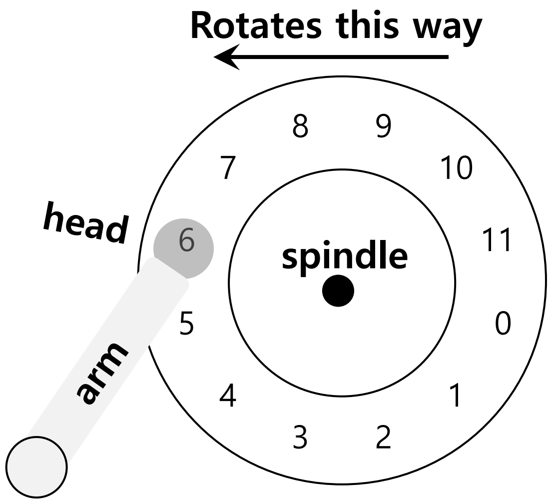

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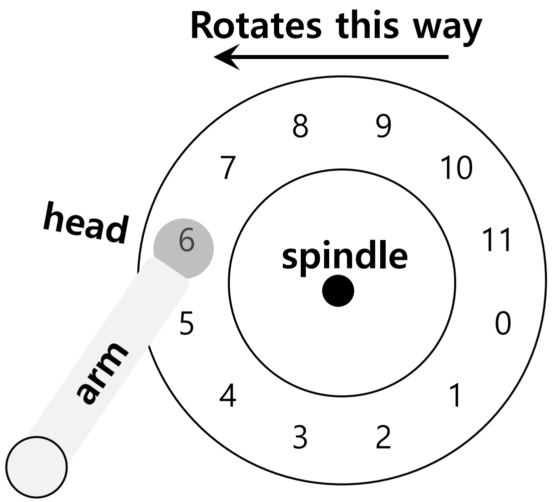

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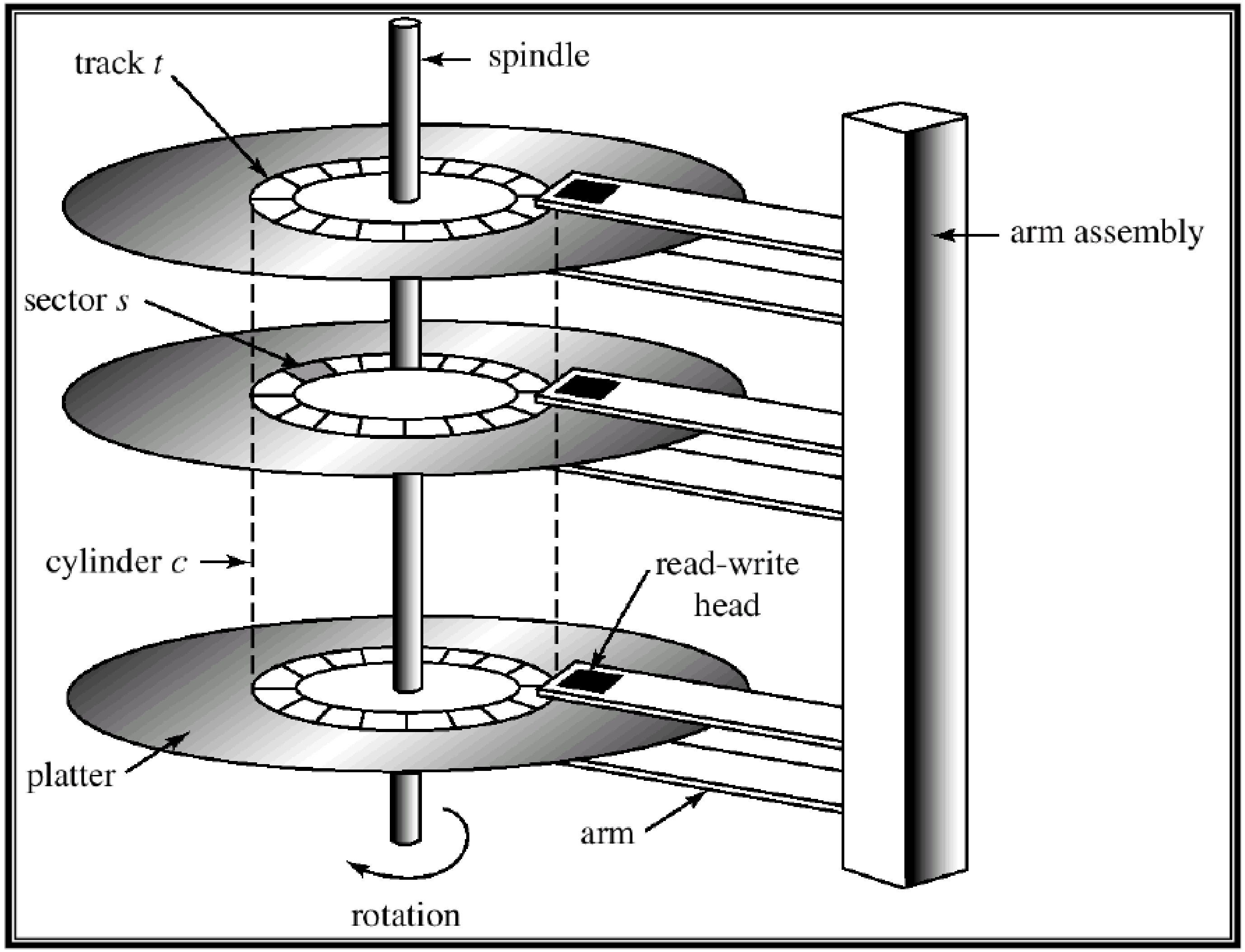
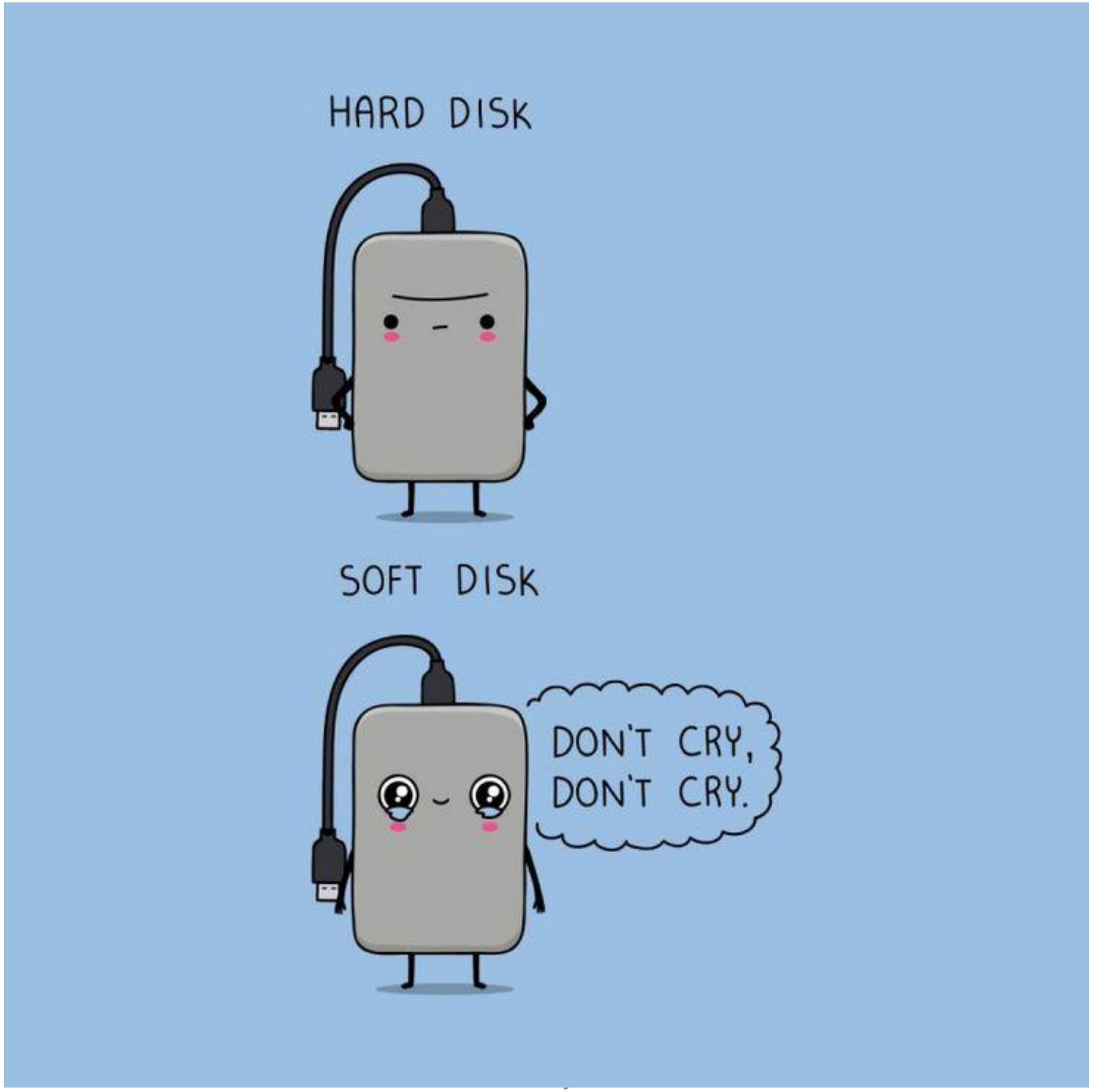
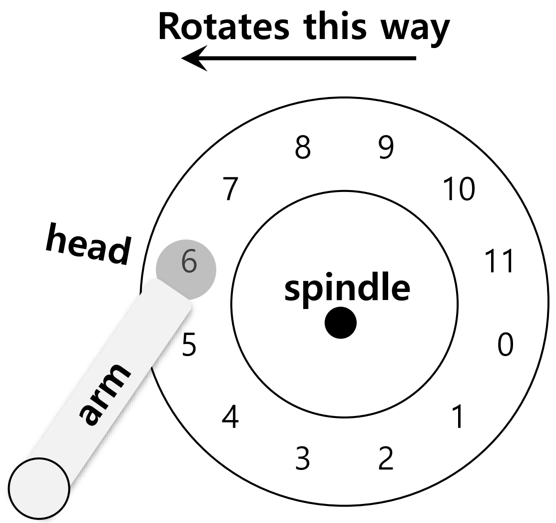



HARD DISK
SOFT DISK
DON'T CRY,
DON'T CRY.

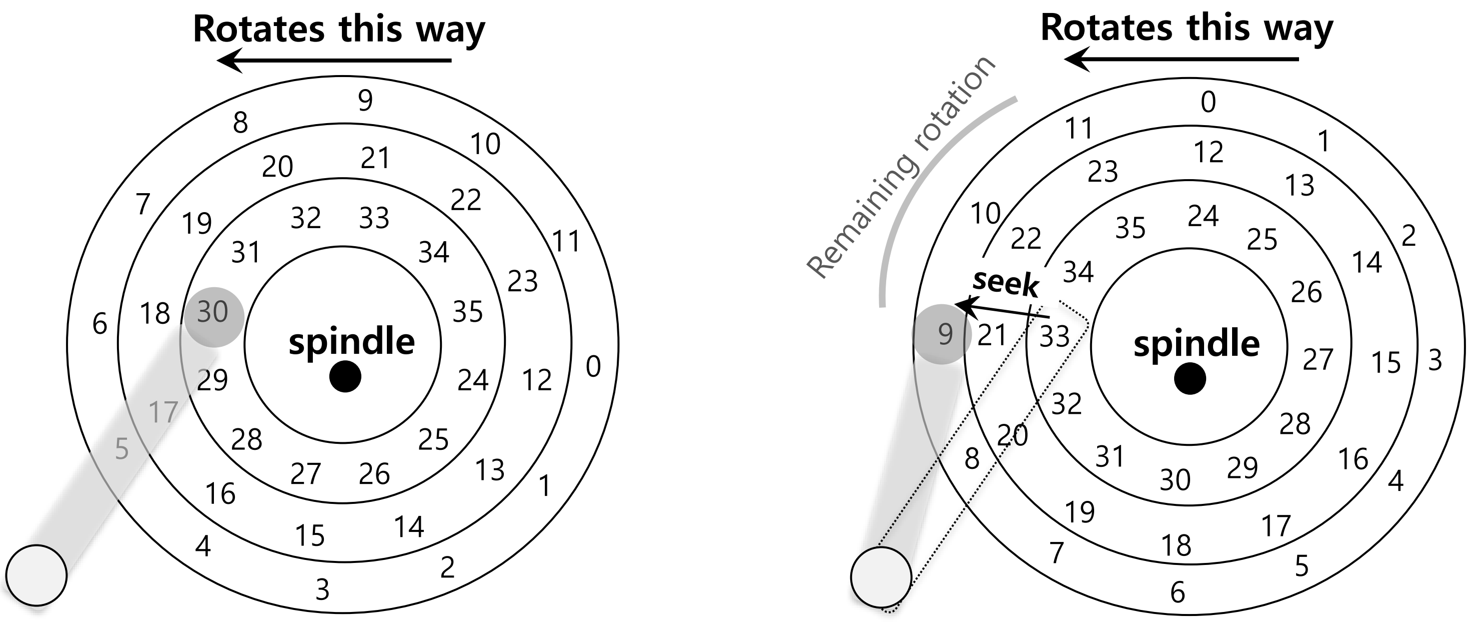
Single-track Latency: The Rotational Delay



A Single Track Plus A Head

- **Rotational delay:** Time for the desired sector to rotate
  - Ex) Full rotational delay is `R` and we start at `sector 6`
    - Read sector 0: Rotational delay = $\frac{R}{2}$
    - Read sector 5: Rotational delay = `R-1` (worst case.)

Multiple Tracks: Seek Time



Three Tracks Plus A Head (Right: With Seek)
(e.g., read to sector 11)

- **Seek**: Move the disk arm to the correct track
  - **Seek time**: Time to move head to the track contain the desired sector.
  - One of the most costly disk operations.

Phases of Seek

- Acceleration → Coasting → Deceleration → Settling
  - **Acceleration**: The disk arm gets moving.
  - **Coasting**: The arm is moving at full speed.
  - **Deceleration**: The arm slows down.
  - **Settling**: The head is *carefully positioned* over the correct track.
    - The settling time is often quite significant, e.g., 0.5 to 2ms.

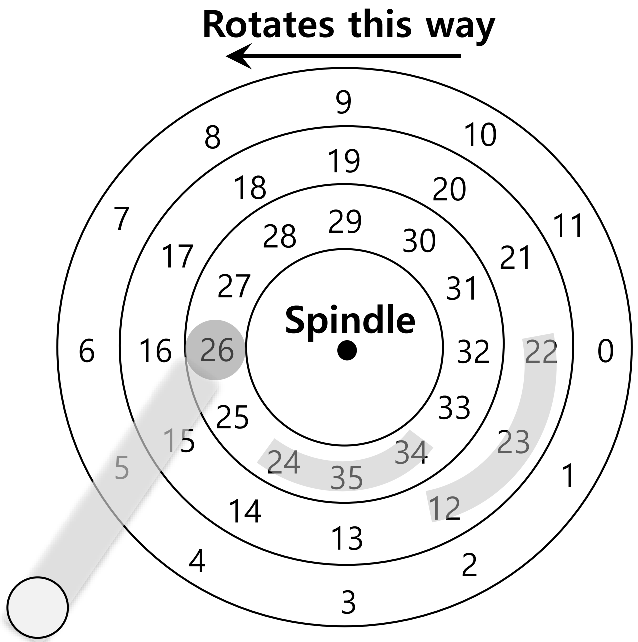
Transfer

- The final phase of I/O
  - Data is either *read from* or *written* to the surface.

- Complete I/O time:
  - **Seek**
  - Waiting for the **rotational delay**
  - **Transfer**

Track Skew

- Make sure that sequential reads can be properly serviced **even when crossing track boundaries**.



Three Tracks: Track Skew Of 2

  - *Without track skew*, the head would be moved to the next track but the desired next block would have already rotated under the head.

Cache (Track Buffer)

- **Hold data** read from or written to the disk
  - Allow the drive to quickly respond to requests.
  - Small amount of memory (usually around 8 or 16 MB)

Write on cache

- **Writeback** (Immediate reporting)
  - Acknowledge a write has completed when it has **put the data in its memory**.
  - faster but dangerous

- **Write through**
  - Acknowledge a write has completed after the write has **actually been written to disk**.

I/O Time: Doing The Math

- I/O time ($T_{I/O}$): $$T_{I/O} = T_{seek} + T_{rotation} + T_{transfer}$$

- The rate of I/O ($R_{I/O}$): $$R_{I/O} = \frac{Size_{Transfer}}{T_{I/O}}$$

	Cheetah 15K.5	Barracuda
Capacity	300 GB	1 TB
RPM	15,000	7,200
Average Seek	4 ms	9 ms
Max Transfer	125 MB/s	105 MB/s
Platters	4	4
Cache	16 MB	16/32 MB
Connects Via	SCSI	SATA

Disk Drive Specs: SCSI Versus SATA

- On the Cheetah:
  - $T_{seek} = \mathbf{4\ ms}$
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- $T_{rotation} = \frac{1}{250} = 4\ ms$. On average, the disk will encounter half a rotation.
- $\therefore T_{rotation} = \frac{4}{2} = 2\ ms.$

- For Random workload, we will consider 4KB.
- Max. transfer = $125\ MB/s \rightarrow$

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- Max. transfer = $125\frac{MB}{s} \rightarrow 125 \times 1024\frac{KB}{s} = 128{,}000\frac{KB}{s}$
- $\therefore, T_{transfer} = \frac{4}{128{,}000} = 31.25\ \mu s \approx \mathbf{30\ \mu s}$

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- $\mathrm{T_{I/O}} = \mathrm{T_{seek}} + \mathrm{T_{rotation}} + \mathrm{T_{transfer}}$
- $\mathrm{T_{I/O}} = 4\ ms + 2\ ms + 30\ \mu s \approx 6\ ms.$

I/O Time Example

- **Random workload**: Issue 4KB read to random locations on the disk
- **Sequential workload**: Read 100MB consecutively from the disk

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T_{seek}		4 ms	9 ms
$T_{rotation}$		2 ms	4.2 ms
Random	$T_{transfer}$	30 microsecs	38 microsecs
	$T_{I/O}$	**6 ms**	13.2 ms
	$R_{I/O}$	0.66 MB/s	0.31 MB/s
Sequential	$T_{transfer}$	800 ms	950 ms
	$T_{I/O}$	806 ms	963.2 ms
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Disk Drive Performance: SCSI Versus SATA

There is a huge gap in drive performance between random and sequential workloads

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Disk Drive Performance: SCSI Versus SATA

- $\mathrm{T_{I/O}}$ for 4KB = 6 ms
- $\mathrm{R_{I/O}}$ = How many MB get transferred in 1 sec.
- $\mathrm{R_{I/O}} = \frac{4KB}{6\ ms} = \frac{4KB}{6} \times 1000\ s \approx \mathbf{0.66\frac{MB}{s}}$

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- For sequential, we will assume a transfer of 100MB
- $\therefore T_{transfer} = \frac{Size_{Transfer}}{R_{I/O}} = \frac{100MB}{125MB/s} = \mathbf{0.8\ s = 800\ ms}$

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$$R_{I/O} = \frac{Size_{Transfer}}{T_{I/O}}$$

I/O Time: Doing The Math

- $T_{I/O} = T_{seek} + T_{rotation} + T_{transfer}$
- $T_{transfer} = 800\ ms$
- $T_{Seek} = 4\ ms$
- $T_{Rotation} = 2\ ms$
- $T_{I/O} = 800 + 4 + 2 = \mathbf{806\ ms}$

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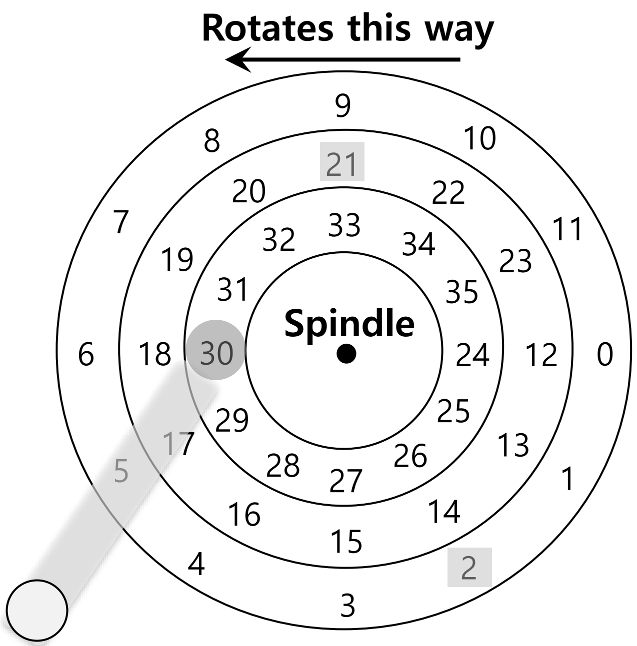
Disk Scheduling

- **Disk Scheduler** decides which I/O request to schedule next.
- **SSTF** (Shortest Seek Time First)
  - Order the queue of I/O request by track
  - Pick requests on the nearest track to complete first



SSTF: Scheduling Request 21 and 2

Issue the request to 21 → issue the request to 2

SSTF is not a panacea.

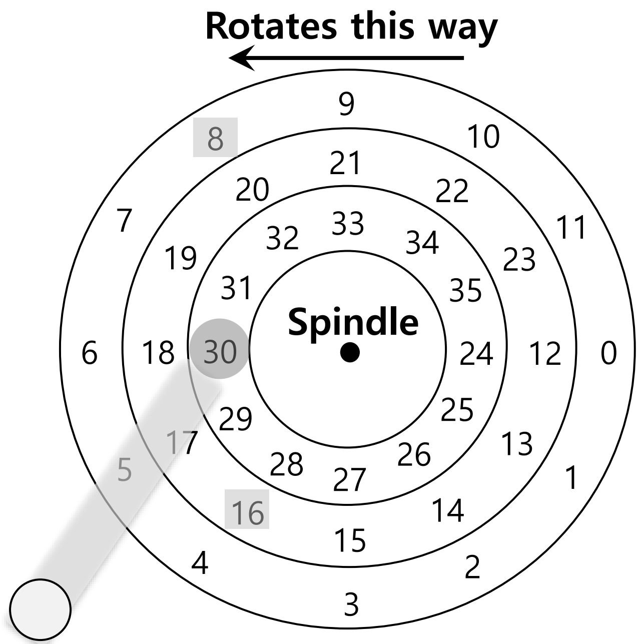
- **Problem 1**: The drive geometry is not available to the host OS
  - Solution: OS can simply implement Nearest-block-first (NBF)

- **Problem 2**: Starvation
  - If there were a steady stream of request to the inner track, request to other tracks would then be ignored completely.

Elevator (a.k.a. SCAN or C-SCAN)

- Move across the disk servicing requests in order across the tracks.
  - **Sweep**: A single pass across the disk
    - If a request comes for a block on a track that has already been services on this sweep of the disk, it is queued until the next sweep.
  - **F-SCAN**
    - Freeze the queue to be serviced when it is doing a sweep
    - Avoid starvation of far-away requests
  - **C-SCAN** (Circular SCAN)
    - Sweep from outer-to-inner, and then inner-to-outer, etc.

How to account for Disk rotation costs?



SSTF: Sometimes Not Good Enough

- If rotation is faster than seek : request 16 → request 8
- If seek is faster than rotation : request 8 → request 16

On modern drives, both seek and rotation are roughly equivalent:
Thus, SPTF (Shortest Positioning Time First) is useful.

I/O merging

- **Reduce the number of request** sent to the disk and lowers overhead
  - E.g., read blocks 33, then 8, then 34:
    - The scheduler merge the request for blocks 33 and 34 *into a single two-block request.*

Conclusion

- Dr. B. Wajid
- Monday: Sep. 2, 2024